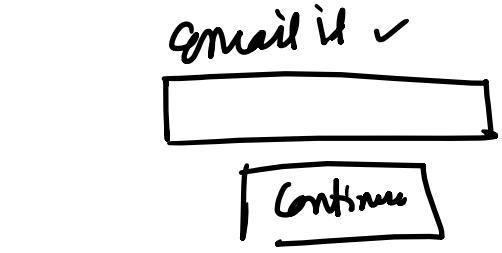
→ Agentic AI foundation (1 Day)
Developing Generative AI Applications on AWS (2 Days)
Building Agentic AI Solutions on AWS (1 Day)
Intermediate

VitalSource



→ Continue without login -

Enterprise customer service system

1. Billing -
2. Technical -
3. Product Recommendation -
4. Refund Requests

1. Imp details may be lost
2. IRRELEVANT Info can distract the model from the current task.
3. Larger Prompts → Token processing, can raise Latency & Inference cost.

(3) Larger prompts → [illegible] & Inference cost.

A

AI Assit (summarize meeting note)

NO

B

AI Sol Analysis.

CT events,
Network conf.
IAM polic
App log.

C

Q&A → Internal HR KB.

Pattern	Simple Analogy	Best suited for
Framwork Layer	→ workflow engine	Controlled coordination
✓ Agent as tool (MCP) ✓	Manager invoking specialists	Reusable capabilities
Agent to Agent (A2A)	Independent services collaborating	Cross-system

Agentcore Memory → E-com DB

Info	
(1) Communication Preferences	→ AgentCore Memory — long term
(2) Conversation events	→ AgentCore Short-term memory
(3) Relevant customer context summaries	→ AgentCore Long-term Memory
(4) order status, Trans details	→ DB / Authorized system

Multi-agents ≠ single Agent

(1) Why do need Multi-agent ?
(2) How do multiple agents communicate & Collaborate ?
(3) How do manage memory & State — multi-agents ?

Lab — Yes / No

"Single-Agent Limitations"

(1) Cognitive overload / load limitations ✓
(2) Context Management / Content window exhaustion
(3) Domain Specialization

Agent mixes Billing procedures — Technical

"Multi-Agent" Communication Patterns

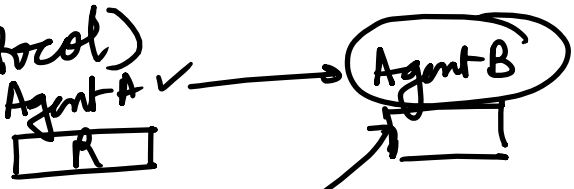
"STANDS"

Multi-Agent ...

① Framework-Layer = "STANDS"

② Agent-as-a-tool

③ Agent-to-Agent (A2A Protocol)



→ Agent Card
→ A2A Client → Communicates
→ A2A Server → Receive / Process the req
→ Task object → work
→ Artifact → output produced

Memory & State Management → AgentCore Memory

Short-term | Long-term

① Short-term memory = Conversational events — interactions & sessions

② Long-term = Preserve useful info — user preferences, semantic facts, summaries -

Activity

① Write → External memory
② Select → Intelligent retrival
③ Compress → Summarize
④ Isolate

Lab:
1. Multi-agents → Strands Agents SDK
2. AgentCore "Runtime"
3. AgentCore "Memory" + AgentCore Identity (Cognito)
4. AgentCore Gateway (Tools)

MCP (Model Context Protocol)

↳ Exposing Tools:

2LO ⇒ service-to-service

3LO ⇒ user-delegated